

# ALBI ZHAKU

US Citizen | az372@njit.edu | 609-686-0252 | LinkedIn: albizh | GitHub: AlbiXD | albizh.com

## EDUCATION

---

### New Jersey Institute of Technology

Newark, NJ

*B.S. in Computer Science*

Expected: May 2027

- **Relevant Coursework:** Operating Systems, GPU Cluster Programming, Intensive Programming in Linux, Computer Networks, Database System Design & Management, Data Structures & Algorithms, Intro to Computer Systems, Intro to Data Science

## PROJECTS

---

### Distributed GPU Video Processor | C++, CUDA, MPI, OpenCV

Dec 2025

- Collaborated in a 2-person team to build a distributed video processor using MPI and CUDA
- Implemented manual frame partitioning and data distribution across MPI processes
- Developed CUDA kernels for per-pixel operations (grayscale, edge-enhance) with optimized thread/block scheduling
- Benchmarked execution modes on a 6,889-frame video: 148s serial CPU, 68s MPI CPU (2.2× faster), and 18s CUDA (8.2× faster)
- Integrated CUDA with MPI and analyzed hybrid performance; observed 5–20% overhead from communication and synchronization during distributed GPU execution

### Minecraft Beta 1.7.3 Server (mineserver-c) | C, epoll, Unix

May 2025

- Built a Minecraft Beta 1.7.3-compatible multiplayer server from scratch in C using non-blocking sockets and `poll`
- Implemented a server-authoritative 20 TPS tick loop with deterministic player state updates and real-time movement synchronization
- Designed a minimal world system with flat chunk transmission, enabling multiple players to spawn and interact in shared space
- Architected an event-driven I/O model theoretically capable of supporting up to ~128 concurrent players on a single instance

### CHIP-8 Emulator | C++, SDL

Dec 2024

- Built a full CPU interpreter implementing 35+ opcodes with accurate memory and stack management
- Integrated SDL graphics and input pipeline to run 20+ ROMs at 60Hz with consistent timing
- Optimized the instruction cycle for stable cross-platform performance

## EXPERIENCE

---

### Forte Pizzeria & Ristorante

Aug 2017 – Present

*Shift Lead / Front Counter Associate, Randolph, NJ*

- Supervised 4-6 staff per shift, coordinated front operations, and trained new employees on service and POS systems, resulting in improved team efficiency and customer satisfaction
- Maintain efficiency and accuracy under peak hours, ensuring positive customer experience and timely order flow

## SKILLS

---

**Programming:** C, C++, Python, Bash, Java, JavaScript, PHP, x86-64 Assembly

**Frameworks & Libraries:** FastAPI, Express.js, Node.js, React.js, SDL, Pandas, NumPy, BeautifulSoup

**Tools & Systems:** CUDA, MPI, Git, Linux, GCC, GDB, Make, Vim, MongoDB, MySQL